Comparison of Near-Infrared and Raman Spectroscopy for the Determination of Chemical and Physical Properties of Naphtha

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Near-infrared (NIR) and Fourier transform (FT) Raman spectroscopy have been compared and evaluated for the quantitative analysis of naphtha with the use of partial least-squares (PLS) regression. Naphtha has been chosen for this purpose since it is a complex mixture of similar hydrocarbons; therefore, both spectroscopic methods can be evaluated with a complex sample of different concentration ranges and also a physical property. Six different chemical compositions [total paraffin, total naphthene (cycloalkane), total aromatic, C6 paraffin, benzene, and cyclopentane] and specific gravity (as a physical parameter) have been selected to evaluate both spectroscopic methods. PLS calibration models for each property have been developed by using both NIR and Raman spectra without spectral pretreatment. Both methods showed good correlation with the corresponding reference methods, but NIR provided the better calibration performance over Raman. The superior signal-to-noise ratio as well as spectral reproducibility of NIR led to the improved calibration performance even though Raman spectroscopy provided richer spectral information. The signal-to-noise ratio, reproducibility of measurement, and richness of spectral information should be simultaneously considered for proper selection of a spectroscopic method, especially for quantitative analysis.

Index Headings: **Near-infrared spectroscopy; Near-IR; NIR; FT-Raman spectroscopy; Partial least-squares regression; Naphtha.**

INTRODUCTION

Near-infrared (NIR) spectroscopy[1] has been widely applied to the monitoring of chemical composition and physical properties of refinery and petrochemical products such as gasoline,[2–3] diesel,[4] para-xylene,[5] and various polymers.[6,7] With NIR spectroscopy, samples can be easily and remotely measured by using optical fibers with good reproducibility. The relevant quantitative spectral information can be successfully extracted with the use of multivariate calibration techniques[8] such as partial least-squares (PLS) regression. Although absorption bands in the NIR region are broad and highly overlapping due to the inherent characteristics of overtone and combination bands, the necessary vibrational information is successfully correlated to sample properties with the use of multivariate calibration techniques. With the help of fast and precise NIR spectroscopic analysis, excellent process control, which more accurately reflects the status of the process at any given time, can be accomplished. Additionally, NIR enables the measurement of multiple properties simultaneously; therefore, investment and operating costs for this system are much more cost-effective compared to the use of several separate conventional analyzers for each property. However, spectral information in the NIR region is limited to mostly C–H, O–H, and N–H vibrations, and it is difficult to use for the measurement of components in very low concentrations because of its inherent low sensitivity. Even with the successful applications of NIR in many fields, there is a reasonable demand for alternative spectroscopic methods that provide more qualitative information, better sensitivity, long-term stability, and on-line capability to overcome the existing drawbacks of NIR spectroscopy.

Raman spectroscopy has recently been emerging as a feasible quantitative analytical method, especially in refinery and petrochemical fields.[9–11] Previously, Raman had been mostly confined to qualitative analyses such as structural identification, while quantitative analyses were performed with a simple band intensity ratio. This is because the Raman scattering intensity is a weak signal and the reproducibility of a Raman spectrum is degraded because of the variation in the excitation laser intensity or changes in the sample matrix. However, due to improvements in laser technology and other instrumental improvements, Raman spectra are now much more stable, and quantitative analysis is possible for complex mixtures, especially when combined with multivariate regression methods. In particular, FT (Fourier transform)-Raman spectroscopy,[12] which utilizes an NIR laser as an excitation source, can be employed for on-line process analysis because it can be easily interfaced with optical fibers to remotely guide the excitation and scattered radiation.[13,14] Additionally, the fluorescence problem commonly seen with visible lasers can be greatly diminished by using a low-energy (longer wavelength) excitation source. Several investigations of FT-Raman spectroscopy for the analysis of products in the refinery and petrochemical areas have been reported.[15,16] Recently, several studies have attempted to compare the performance of NIR, mid-IR, and Raman spectroscopies for the determination of BTEX (benzene, toluene, ethylbenzene, xylene)[17] and percent oxygen in petroleum fuels.[18,19] However, those comparisons were based on the determination of only a few specific components instead of many components with different concentration ranges as well as physical properties.

The major goal of this study is to evaluate both NIR and FT-Raman spectroscopy for the quantitative analysis of naphtha combined with PLS regression and to rationalize important spectral parameters influencing the performance of quantitative calibration. Both spectroscopic methods operate in the NIR region; however, each has independent advantages and disadvantages that need to be considered for quantitative analysis. In order to deter-

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mine which technique and which spectral parameters are the most important for quantitative determinations in the petrochemical industries, a naphtha matrix was chosen for chemical composition and physical property measurements. Naphtha was selected for this study since it is a complex mixture of similar hydrocarbons and there are many different chemical and physical properties to be monitored. Therefore, both spectroscopic methods can be evaluated in a complex sample matrix with different concentration ranges (0.1 to over 50 vol %) as well as a physical property. Six different chemical compositions of total paraffin, total aromatic, total naphthene, C6 paraffin, benzene, and cyclopentane in naphtha were selected, and specific gravity was chosen as a physical property.

PLS calibration models for each property were developed by using both NIR and Raman spectra without spectral pretreatments. Both methods showed good correlation with reference methods. However, NIR provided the better correlation and predicted error. The superior signal-to-noise ratio and spectral reproducibility of NIR led to the improved calibration performance, even though Raman spectroscopy provided higher spectral resolution. Along with the spectral resolution, the signal-to-noise ratio and reproducibility of measurement should be seriously considered in order to achieve the optimum performance of a quantitative analysis.

EXPERIMENTAL

Instrument and Apparatus. *NIR Spectra.* NIR spectra were collected over the 1100 to 2500 nm spectral region with a NIRSystems Model 6500 spectrometer (Foss NIRSystems, Silver Spring, MD) equipped with a tungsten–halogen lamp, PbS detector, and fiber-optic interactance/reflectance probe. The resolution of the collected spectra was 10 nm with 2 nm data intervals. The fiber-optic probe consisted of concentric rings of illuminating fibers (inner core, 210 fibers), receiving fibers (outer ring, 210 fibers), and a reflecting mirror. The size of the fiber-optic probe was 2.54 cm outer diameter and 15.2 cm in length. The length of the optical fibers from the probe to spectrometer was 2 m. The distance between the optical fibers and the reflecting mirror was 0.5 cm, resulting in an actual pathlength of 1 cm. NIR spectra were collected by positioning the fiber-optic probe into each sample that was held in a sealed bottle. Each NIR sample spectrum consisted of 16 coadded scans (acquisition time: 10 s).

Raman Spectra. FT-Raman spectra were collected on a Bruker IFS55 FT-IR spectrometer equipped with a Bruker FRA 106 Raman module, CaF_2 beamsplitter, and cryogenically cooled Ge detector. Both Stokes and anti-Stokes spectral regions were collected. The excitation source used was a diode-pumped Nd:YAG laser operated at 1064 nm with an output power of 200 mW. The Rayleigh line was rejected by using a dielectric filter disk. The sample was transferred in a rectangular cuvette with back-reflecting mirror, and 128 scans (acquisition time: 220 s) were taken at a resolution of 4 cm^{-1} in a 180° scattering arrangement. Prior to Fourier transformation, the interferograms were apodized with a Blackman–Harris four-point filter and zero-filled. All the Raman and NIR spectra were collected at room temperature (23 °C).

Procedures. *Sample Preparation.* Fifty naphtha samples were obtained over a 6 month period at SK Corporation (Ulsan, Korea). Over this long period, samples were carefully collected to contribute more concentration- and process-related variations to the data set, which would result in more robust calibration models. The average concentrations of each component widely varied from 0.10 to 54.67 vol %. Immediately after collection, samples were sealed and stored in a refrigerator at 4 °C to prevent evaporation of the hydrocarbons. The concentrations of each sample were measured by a Hewlett-Packard 5890 GC (Reformulyzer) equipped with three separation columns and a flame ionization detector (FID). Specific gravity of the samples was determined by a conventional density meter at 15.6 °C.

PLS Regression. PLS regression was performed with the use of GRAMS/32 software with add-on PLS algorithm (Galactic Industries Corporation, Salem, NH). NIR and Raman spectra were imported into GRAMS/32 before the PLS regression was accomplished. A total of 50 spectra were divided into 35 spectra for calibration and 15 spectra for the prediction data set, which served as a validation set. Spectra in the calibration and prediction set were randomly chosen.

RESULTS AND DISCUSSION

Spectral Features. For the selective and simultaneous quantification of components in a complex mixture such as naphtha by spectroscopic methods, it is required that the minute spectral differences among components be examined. For qualitative examination of spectral features of typical representative components in naphtha, both NIR and Raman spectra of *n*-hexane, cyclohexane, toluene, and naphtha are shown in Fig. 1. All spectra were linearly offset for the purpose of qualitative comparison.

In the NIR spectra, the most useful spectral information is located in the 1100–1650 and 1800–2200 nm spectral regions. The 1650–1800 and 2200–2500 nm regions contain no useful spectral information, because of the strong and saturated absorption of the NIR radiation from the long optical pathlength. The bands around 1200 nm correspond to the second overtone of C–H bands from the 3100–2700 cm^{-1} mid-infrared region (e.g., 2870 $cm^{-1} \times 3 = 8610\ cm^{-1} = 1161$ nm) as shown in the Raman spectra. The bands around 1400 nm correspond to the combination bands between the first overtone of C–H stretching from the 3100–2700 cm^{-1} region and the CH_3/CH_2 bending around 1450 cm^{-1} (e.g., 2870 $cm^{-1} \times 2 + 1450\ cm^{-1} = 7190\ cm^{-1} = 1390$ nm). The absorption bands in the 1800–2200 nm region correspond to the shoulder of the strongly absorption combination band around 2300 nm, which is due to the coupling between the C–H stretching band in the 3100–2700 cm^{-1} region and CH_3/CH_2 bending band around 1450 cm^{-1} (e.g., 2870 $cm^{-1} + 1450\ cm^{-1} = 4320\ cm^{-1} = 2314$ nm). The spectral variations in the 1800–2200 nm range are more dominant than those in the 1100–1650 nm region. The spectral features of toluene are significantly different from those of aliphatic hydrocarbons because of the clear structural difference. In toluene, the second overtone band of C–H stretch (1140 nm) attached to the benzene ring has been clearly observed. *n*-Hexane and cyclohexane also show noticeable spectral differences in the

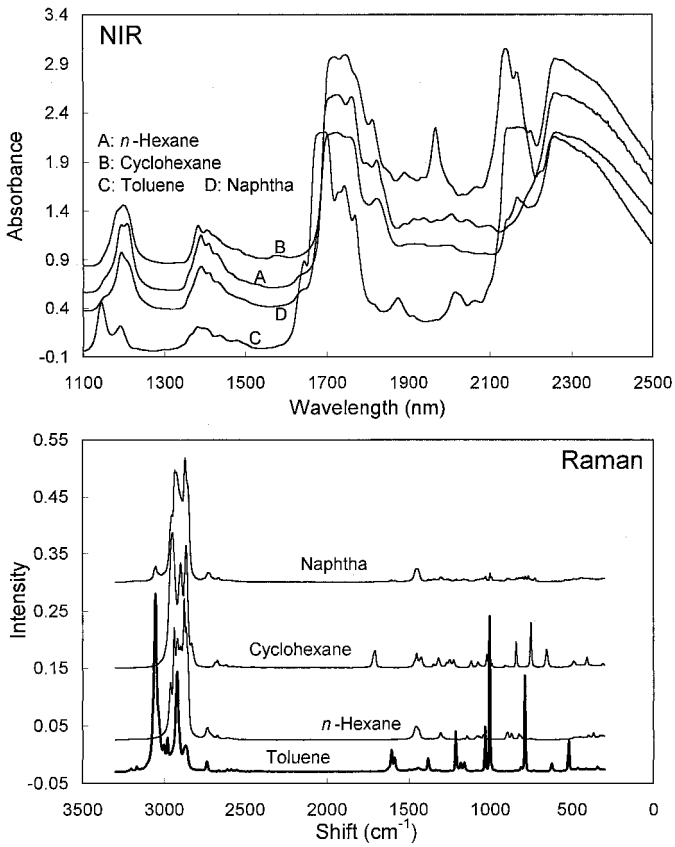

FIG. 1. NIR and Raman spectra of *n*-hexane, cyclohexane, toluene, and naphtha (offset for clarity).

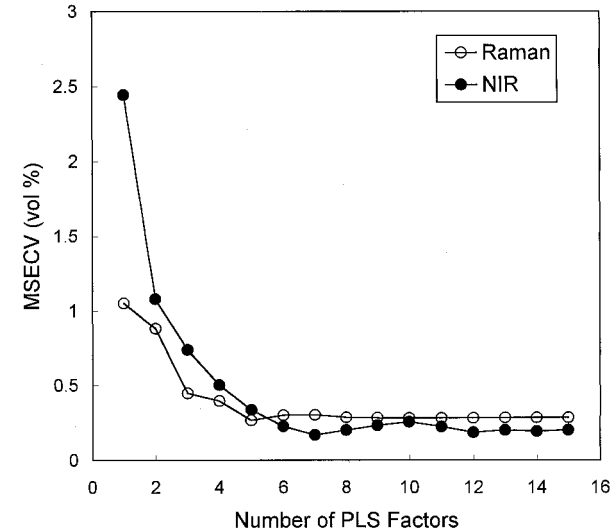

FIG. 2. The effect of number of PLS factors on MSECV for total paraffin content determination within the 1100–1650 nm range for NIR and 3200–2500 cm^{-1} for Raman.

1800–2200 nm region. The NIR spectral features of components in Fig. 1 are very broad compared to those in the Raman and in the mid-infrared region; however, the spectral differences are noticeable. Consequently, there are distinct spectral differences among paraffin, naphthene, and aromatic hydrocarbons in the NIR region.

The Raman spectra are much clearer, and sharper spectral differences among the components are observed. The most useful information is located in the 3200–2500 and 1700–300 cm^{-1} regions. The bands at 3200–2500 and 1700–300 cm^{-1} correspond to C–H stretching and bending, respectively. For toluene, intense and sharp C–H bands attached to the benzene ring are noticeably different from aliphatic C–H bands, which are clearly observed at 3050 and 1650 cm^{-1}. In addition, spectral features in the 1700 to 300 cm^{-1} region are clearly different from those of aliphatic hydrocarbons. For cyclohexane, overall spectral features are unique and different, especially the band at 750 cm^{-1} due to the breathing motion of the hydrocarbon ring. The spectral features of *n*-hexane are also quite different from those of other components. The overall Raman spectral features of each component are very clear and easily identified.

Overall, Raman spectra provide much better qualitative and structural information compared to NIR spectra. Conversely, it has been reported that NIR yields a better signal-to-noise ratio over Raman.[17] When both spectroscopic methods are used for the quantitative analysis of very complex mixtures such as naphtha, which is composed of more than 100 individual components and various kinds of hydrocarbons with similar structures, the resulting spectrum will be highly overlapped regardless of spectral bandwidth. NIR and Raman spectra of typical naphtha are also shown in Fig. 1, and the resulting spectra are overlapped with each of the components in the mixture. Especially, in the 1700 to 690 cm^{-1} region in the Raman spectrum, we see that the sharp peaks of individual components are overlapped and the resulting peaks are relatively broad. The aromatic C–H band at 3050 cm^{-1} is still clearly identified. Both NIR and Raman spectra of naphtha are highly overlapped; therefore, multivariate calibration methods are essential to resolve these mixtures, as has been determined empirically in the field.

PLS Calibration Models. In the application of the PLS algorithm, it is generally known that the spectral range and number of PLS factors are critical parameters. It has been previously determined that the calibration performance depends on the spectral region utilized.[5] For NIR, three different spectral regions—1100–1650, 1800–2200, and the combination of 1100–1650 and 1800–2200 nm—were examined. These spectral regions were selected to include combinations of the two major absorption areas (1100–1650 and 1800–2200 nm). Similarly, three different spectral regions—3200–2500, 1700–190, and the combination of 3200–2500 and 1700–190 cm^{-1}—were examined for Raman. These spectral regions were selected to include combinations of the two major scattering areas (3200–2500 and 1700–190 cm^{-1}).

The optimum number of factors was identified as the number of factors that gives a minimum MSECV (mean square error of cross-validation). The cross-validation was used as a validation method by dividing the calibration data set into seven segments. The MSECV is a good index to find the optimal number of PLS factors without overmodeling. Figure 2 shows the MSECV plotted as a function of the number of PLS factors used to determine the total paraffin content within the 1100–1650 nm range for NIR and 3200–2500 cm^{-1} for Raman. As expected, the MSECV decreases sharply for the initial factors and gradually decreases as more paraffin-dependent spectral

TABLE I. Best PLS calibration models for each component with the use of NIR and Raman spectroscopy.

	NIR				Raman			
Component	Number of factors	Spectral range (nm)	MSECV	SEP	Number of factors	Spectral range (cm^{-1})	MSECV	SEP
Total paraffin (vol %)	7	1100–1650	0.169	0.092	5	3200–2500	0.265	0.184
Total naphthene (vol %)	6	1100–1650	0.175	0.108	4	3200–2500	0.230	0.184
Total aromatic (vol %)	7	1100–1650	0.098	0.062	6	3200–2500	0.099	0.062
C6 paraffin (vol %)	9	1100–1650	0.219	0.141	5	3200–2500 & 1700–190	0.271	0.218
Benzene (vol %)	7	1800–2200	0.030	0.018	4	1700–190	0.073	0.036
Cyclopentane (vol %)	9	1100–1650	0.013	0.016	3	3200–2500	0.025	0.195
Specific gravity (10^{-4})	6	1100–1650 & 1800–2200	2.3	2.4	6	3200–2500 & 1700–190	3.4	3.6

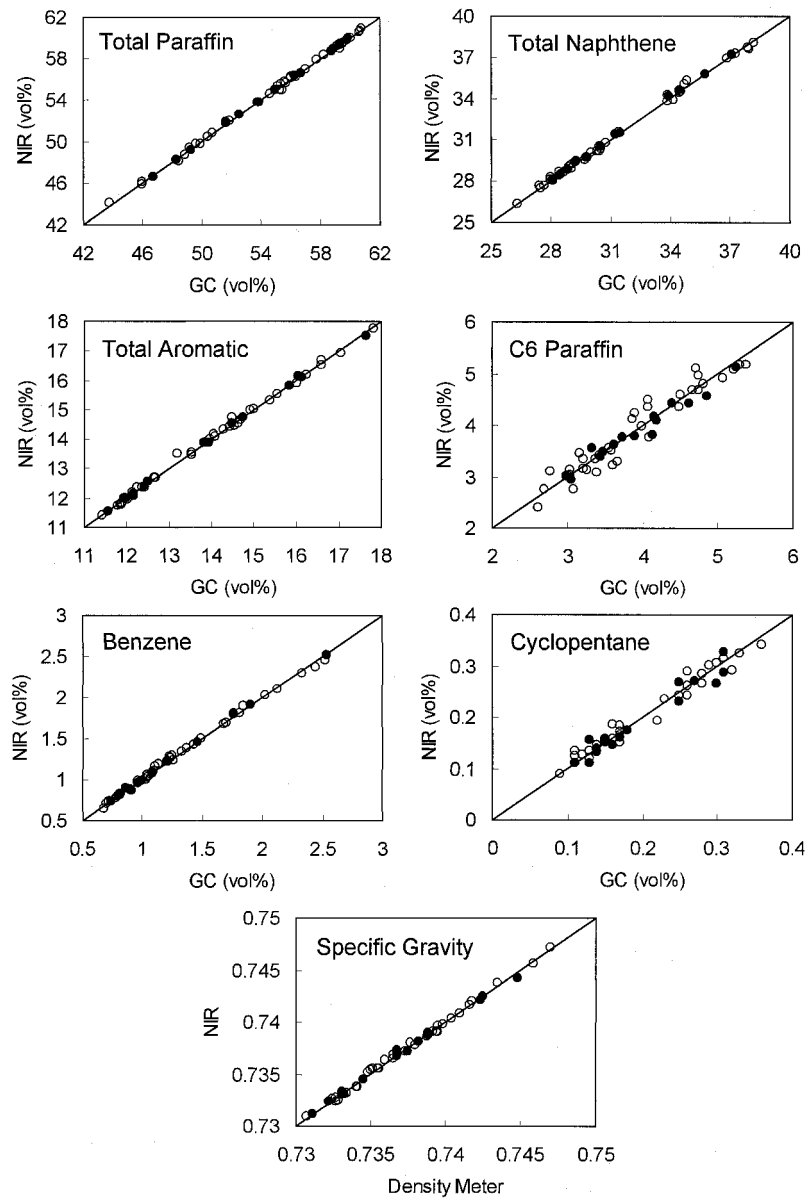

FIG. 3. Scatter plots showing the correlation between NIR and reference analyses for each component. Open and filled circles correspond to calibration and prediction data, respectively.

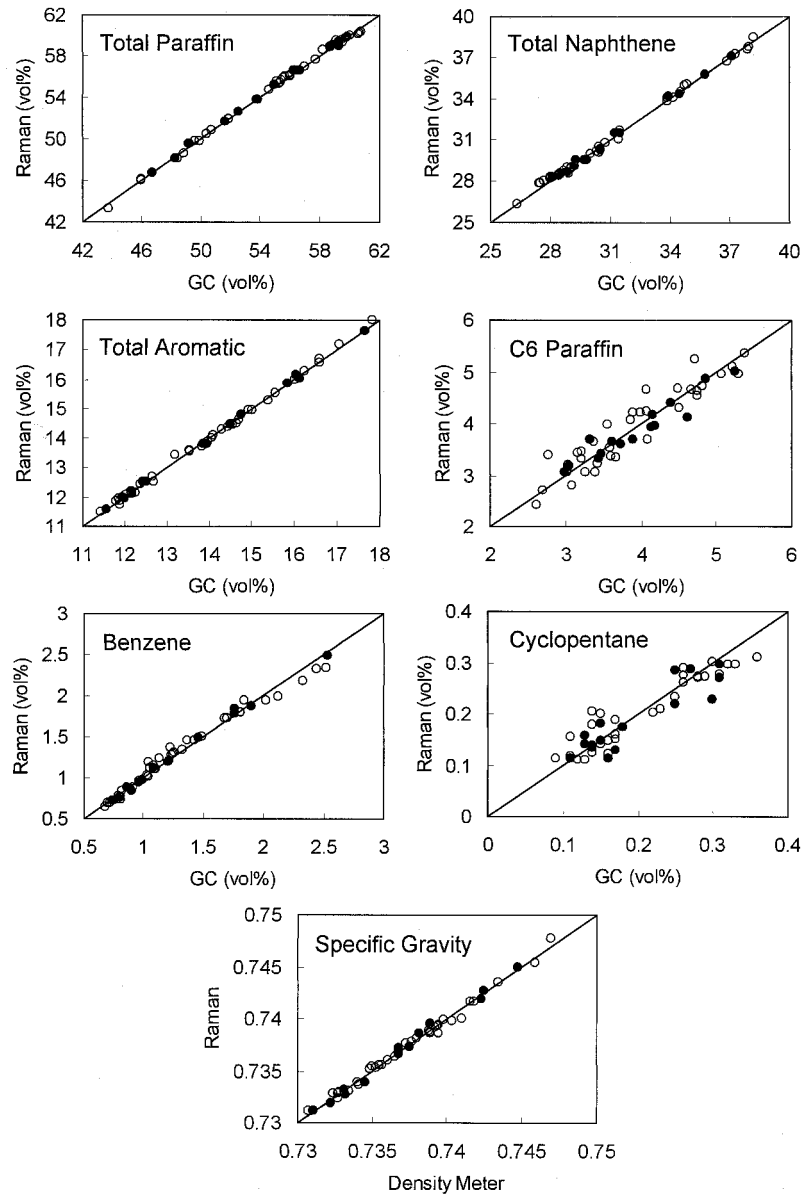

FIG. 4. Scatter plots showing correlation between Raman and reference analyses for each component. Open and filled circles correspond to calibration and prediction data, respectively.

variation is incorporated into the calibration model. The decrease of MSECV is shaper for NIR as compared to Raman. As the number of factors increases further, the MSECV begins to increase at eight and six PLS factors for NIR and Raman, respectively. An increase in the MSECV indicates that the data have been over-fit by incorporating spectral information into the model that is not related to paraffin. In this case, seven and five PLS factors were chosen as the optimum for NIR and Raman, respectively. The optimum number of PLS factors for other spectral regions and other components was determined by generating and examining the same types of plots.

The results of the best calibration model for each component are summarized in Table I. Standard error of prediction (SEP) was calculated by predicting the samples in the prediction data set. As shown in Table I, the calibration and prediction performances for each component with the use of NIR are better than Raman, except for total aromatic content. The calibration results of total aromatic content with the use of both NIR and Raman are similar. As shown in Fig. 1, there is a unique and non-overlapping band for the aromatic compounds at 3050 cm^{-1}, which is noticeably different from aliphatic C–H bands. This unique band provides the improved calibration performance when Raman is used. Overall, NIR requires more PLS factors to achieve optimal calibration models, since the bands in the NIR region are broader and less characterized as compared to Raman. Therefore, more factors are necessary to describe the analyte variations in NIR spectra. In contrast, the sharper and well-characterized Raman bands require fewer numbers of factors to determine related analyte information. Especially for single-component quantification such as that for benzene and cyclopentane, Raman requires significantly fewer factors as compared to NIR, since the spectral features

TABLE II. **Average intensity, standard deviation, signal-to-noise ratio, and relative standard deviation (RSD) of 20 spectra of the same sample in different spectral ranges.**

	Spectral range	Average intensity	Standard deviation	Signal-to-noise ratio	RSD (%)
NIR	1100–1650 nm	0.188671	0.000328	575.2	0.21
	1800–2200 nm	0.845703	0.001352	625.5	0.16
	1100–1650 & 1800–2200 nm	0.465534	0.000760	612.5	0.18
Raman	1700–190 cm^{-1}	0.006512	0.000183	35.6	3.15
	3200–2500 cm^{-1}	0.039411	0.000601	65.6	2.71
	3200–2500 & 1700–190 cm^{-1}	0.016952	0.000316	53.6	3.01

of the components are clearer and sharper in the Raman spectrum. For the calibration of chemical compositions, the optimum spectral regions for each component are different. However, the full spectral range gives the best calibration result for specific gravity both in NIR and Raman. This is because specific gravity, a physical parameter, is the result of the contribution not from specific components but from all components in naphtha. Therefore, to build the optimum calibration model, the full spectral range that contains full spectral features of all components provides the best results, rather than specific spectral regions. Additionally, the required number of PLS factors is the same for both spectroscopic methods, in that the degree of spectral variations related to specific gravity in full spectral range is similar regardless of the characteristics of the NIR and Raman bands. Scatter plots showing correlation between both methods and reference analyses for each component are shown in Figs. 3 (NIR) and 4 (Raman). Open circles and filled circles represent calibration and prediction data, respectively. For all components, both methods have good correlation with corresponding reference methods. However, the NIR calibration and prediction data are closer to the unity line in comparison to the Raman data.

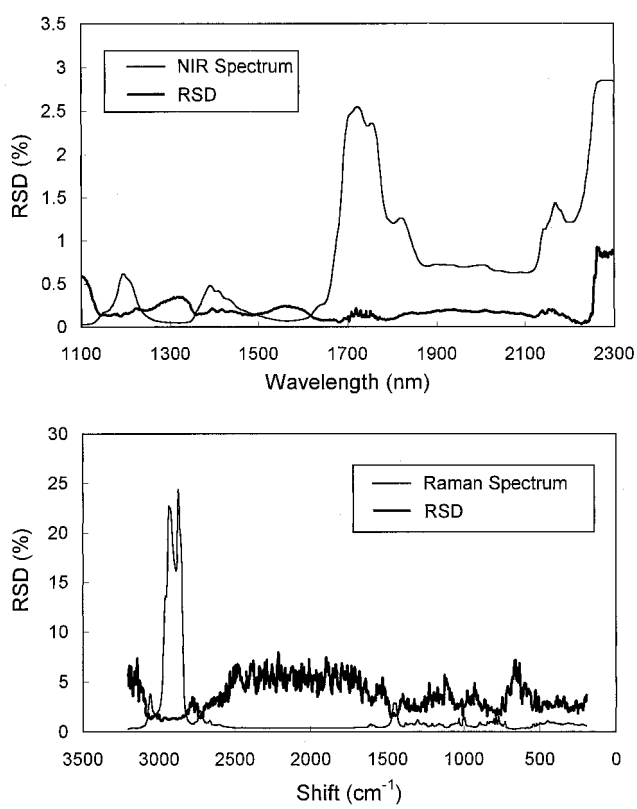

FIG. 5. The relative standard deviations (RSDs) at each wavelength for NIR (top) and Raman (bottom). The naphtha spectrum was scaled for the purpose of comparison.

Signal-to-Noise and Reproducibility Test. To investigate the better calibration performance using NIR, we evaluated the signal-to-noise ratio and spectral reproducibility for both spectroscopic methods. The signal-to-noise ratio and spectral reproducibility are two of the most important parameters in determining the final performance of a quantitative calibration. The signal-to-noise ratio and reproducibility of a spectrum are the result of multiple factors such as the performance of instrumental components, physical status of sample, etc. If a collected spectrum is noisy and not reproducible enough, even though the qualitative information is rich, the result from the quantitative calibration would not be satisfactory even with necessary mathematical pretreatments of the spectrum. To test the signal-to-noise ratio and reproducibility of both methods, we continuously collected 20 NIR and Raman spectra of the same naphtha sample over a 10 h period at 30 min intervals on the same day. For NIR, a background single-beam spectrum of air was collected for each measurement immediately before the collection of the sample single-beam spectrum. For Raman, the sample was kept in the same position during the entire measurement. The sample was tightly sealed to prevent any evaporation during this period.

To evaluate signal-to-noise ratio and spectral reproducibility, we calculated the average intensity (absorbance or scattering) of 20 spectra of the same sample at each wavelength, as well as the corresponding standard deviation and relative standard deviation (RSD). Then, those calculated values at each wavelength were averaged for the same spectral ranges used to build PLS calibration models. The signal-to-noise ratio was calculated by dividing the average intensity by the standard deviation. The RSD was used to evaluate the variations of both spectra. The results are summarized in Table II. As shown in Table II, NIR shows the highest signal-to-noise ratio (approximately 10-fold) compared to the inherently weak Raman scattering. The lower noise and higher intensity of NIR resulted in the improved calibration performance over Raman. The average RSDs of NIR spectra in the spectral ranges used for calibration are superior to those of Raman spectra. The highly absorbing region from 1800 to 2200 nm in the NIR and the Raman scattering

TABLE III. The standard deviations of predicted values from 20 spectra of the same sample with the use of best calibration models for each component.

	Total paraffin (vol %)	Total naphthene (vol %)	Total aromatic (vol %)	C6 paraffin (vol %)	Benzene (vol %)	Cyclopentane (vol %)	Specific gravity
NIR	0.047	0.042	0.020	0.092	0.015	0.004	0.004
Raman	0.162	0.229	0.115	0.182	0.018	0.007	0.005

region at 3200 to 2500 cm^{-1} show better RSDs in comparison to other spectral regions.

The RSDs at each wavelength for NIR and Raman are graphically shown in Fig. 5. Both figures are compared with the average naphtha spectrum. The average naphtha spectrum was scaled for better qualitative comparison. RSD patterns for NIR and Raman are quite different. The lower RSDs were achieved in the highly absorbing and scattering region. In NIR, RSDs near the 1200, 1400, and 1800–2200 regions are fairly similar; however, weakly absorbing regions around 1110, 1300, and 1580 nm show higher RSDs. Similarly in Raman, the highly scattering regions at 3150–2700 and 1480 cm^{-1} show the lower RSDs compared to weak or nonscattering regions at 1400–190 and 2500–1700 cm^{-1}, which show a noisy pattern.

The reproducibility of actual prediction with the use of the best calibration model for each component has been evaluated by calculating the standard deviation of predicted values from 20 spectra collected from the same sample. The results are summarized in Table III. Similar to the reproducibility results of the spectra themselves, the standard deviation of predicted values of NIR shows the best results vs. Raman for all components. The variation of Raman intensity is mostly due to the change of laser intensity, since Raman intensity is subject to excitation intensity. Even though the power of the laser is continuously controlled, minor change in laser intensity affects the intensity of Raman scattering in the whole spectral range. In the NIR measurement, the source intensity variation is continuously being compensated for by the collection of a reference single-beam spectrum. However, Raman is the measurement of absolute scattering intensity. The lower noise level and better reproducibility of the NIR spectrum lead to more reproducible prediction results with PLS calibration models. This result shows that the reproducibility as well as signal-to-noise ratio of a spectrum directly influences the performance of PLS calibration models.

Although NIR spectra are broad and less qualitative, the superior signal-to-noise ratio and reproducibility result in better calibration performance over Raman. The excellent reproducibility of NIR in conjunction with chemometrics makes it possible to apply the techniques to wide areas for process control and optimization. However, NIR is still confined to the analysis of samples with high concentration of analytes, and the information is limited mainly to C–H, N–H, and O–H vibrations. Recently, many research efforts have focused on Raman spectroscopy in conjunction with chemometrics for quantitative analysis. The improvement in the stability and signal-to-noise ratio of the Raman spectrum along with proper use of an external reference will be a great advantage in practical applications, since it could provide richer chemical information over NIR and be easily adapted to the on-line measurement of highly viscous and nontransparent samples.

CONCLUSION

NIR and Raman spectroscopy have been compared for quantitative analysis of the chemical and physical properties of naphtha. NIR was found to have better calibration performance over Raman in this study. Even though Raman has much higher spectral resolution over NIR, the superior signal-to-noise ratio and spectral reproducibility of NIR lead to better calibration results. The reproducibility of the measurement including signal-to-noise ratio and the spectral resolution should be simultaneously considered for the selection of the spectroscopic method for quantitative analysis. Successful implementation of NIR has changed the process control and optimization concept in many refinery and petrochemical industries. As a second runner, FT-Raman spectroscopy has strong potential for on-line quantitative analysis since it provides richer chemical information that can be easily interfaced with optical fiber methods for remote monitoring without sample pretreatment. Considerable effort will need to be made to improve the signal-to-noise ratio, so that the information from Raman spectroscopy demonstrates the precision and reproducibility needed. It is obvious that Raman intensity fluctuates as laser power varies; therefore, Raman spectra should be referenced by using laser output or other references for the quantitative analysis. The reference was not used to correct Raman intensity in this study; however, the proper use of a reference will improve the reproducibility of Raman spectra and quantitative calibration performance. Future work will evaluate remote FT-Raman spectroscopy using optical fibers with a proper external reference for the measurement of important properties in refinery and petrochemical areas.

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